

HUAWEI TECH ARENA 2026 SUB-TOPIC 3

The architecture, component by component

What each part of the system is, why it is there, and what it earns against the two criteria the competition actually marks.

Built and measured on the 540-design dataset supplied by Huawei.

THE PROBLEM

One simulation is cheap. A hundred thousand are not.

Why a simulation is slow, and why the model is not

THE SIMULATOR



THE TRAINED MODEL



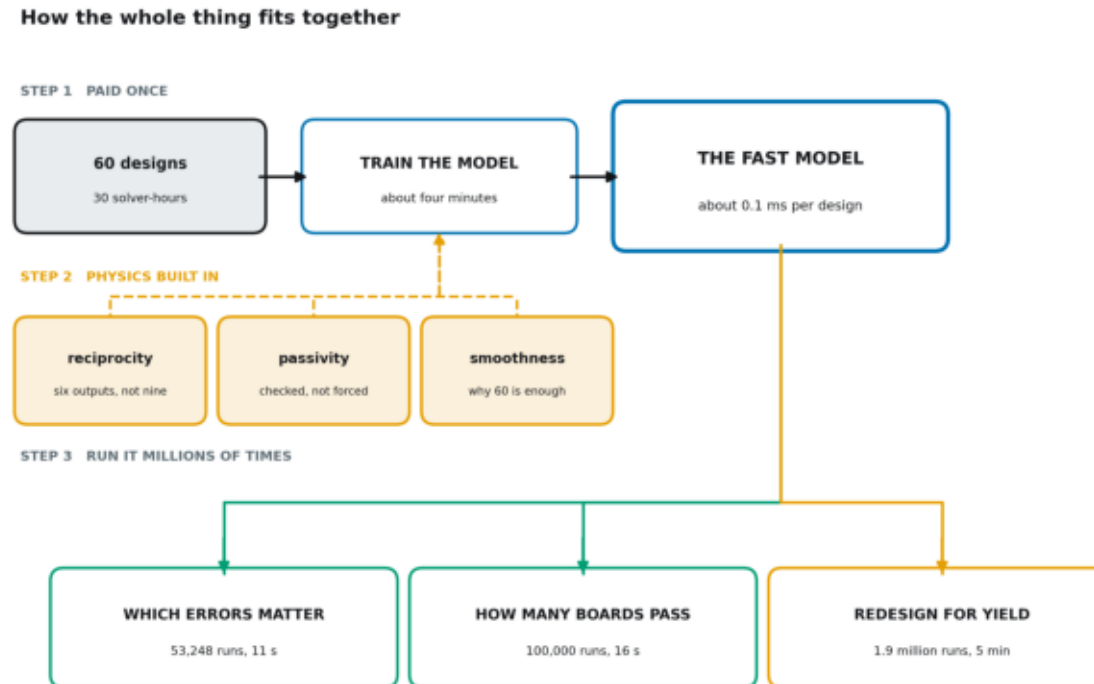
A full-wave simulator solves Maxwell equations for the fields throughout the structure, then reports what comes out of each port. It is accurate and it is slow.

That is fine when you need one answer. It is hopeless when you need to know how a hundred thousand manufactured boards will behave, which is what a tolerance study asks.

The trained model never solves Maxwell. It learned the pattern from examples that did, so it answers in a tenth of a millisecond instead of 30 minutes.

THE SYSTEM

How the whole thing fits together



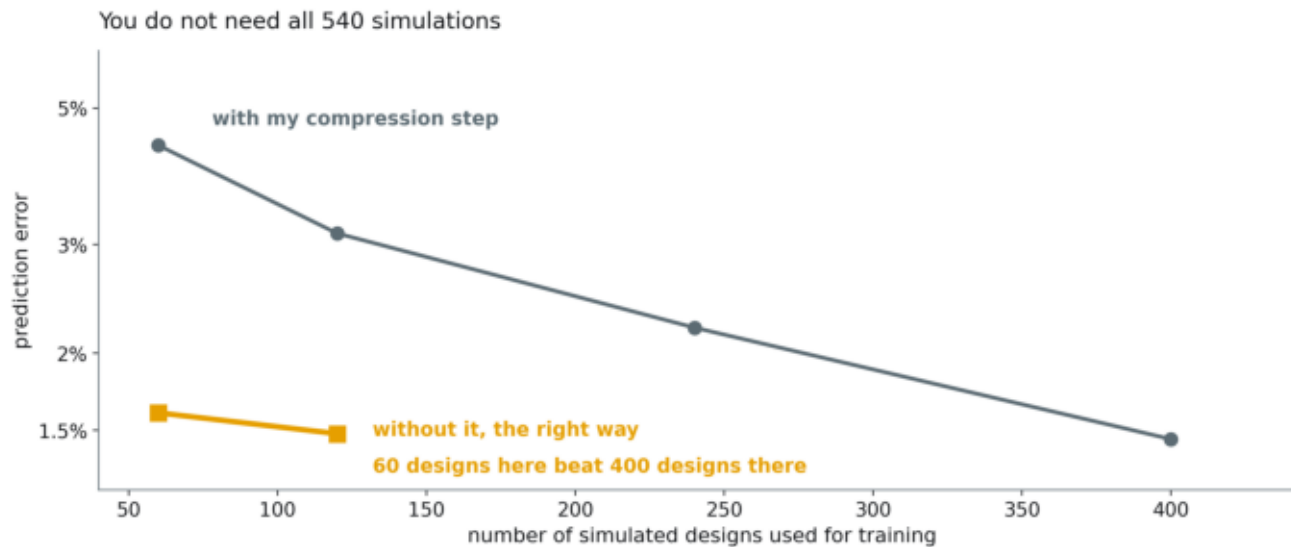
STEP 1 is paid once. Sixty simulations, 30 solver-hours, then about four minutes of training. Everything after is cheap.

STEP 2 is where physics enters. Three laws the measurement already obeys are built in or verified, not learned.

STEP 3 is the payoff. Three studies that were previously unaffordable, each now taking seconds.

COMPONENT 1

The training data, and how little of it you need



WHAT IT IS

Designs simulated in the real solver, each an input and its measured answer.

WHY IT MATTERS

Simulations are the only real cost in the project. Every one you avoid is time and money saved.

WHAT WE MEASURED

Sixty designs reach engineering accuracy. That is one ninth of what Huawei supplied.

The brief names data availability and training cost as two of its four technical challenges. Sixty designs is 30 solver-hours instead of 270, a claim a jury can check.

COMPONENT 2

The physics, and an honest negative result

Three laws your measurements already obey



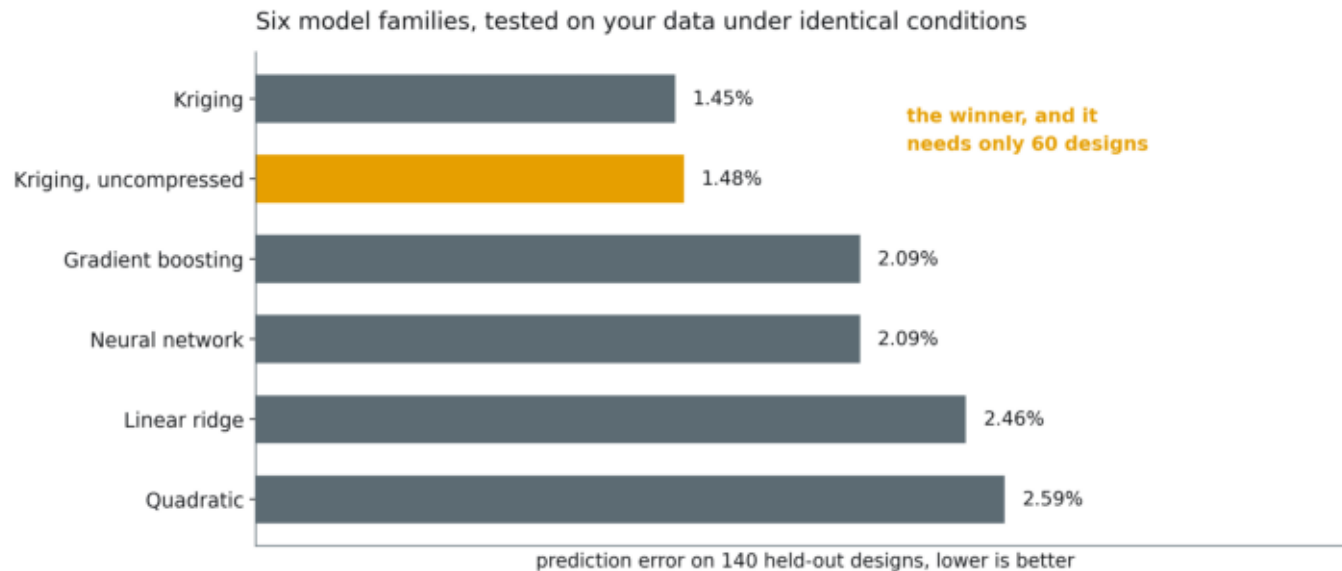
Reciprocity is imposed by construction. The model predicts six numbers and copies the other three, so the law can never be broken and the problem shrinks by a third.

Passivity was tested as a training constraint, as a penalty and as a hard projection. Neither changed the error to four decimals, and both cost thirty times the training time.

Reporting that is worth more than hiding it. A constraint only pays where the model actually breaks it, and this device never came close.

COMPONENT 3

Which model, and why that one



HOW THIS WAS DECIDED

Six families, identical splits, five repeats, 140 designs never seen.

THE SURPRISE

A plain linear fit came within a factor of two of the best model. Any proposal that skips that baseline is exposed.

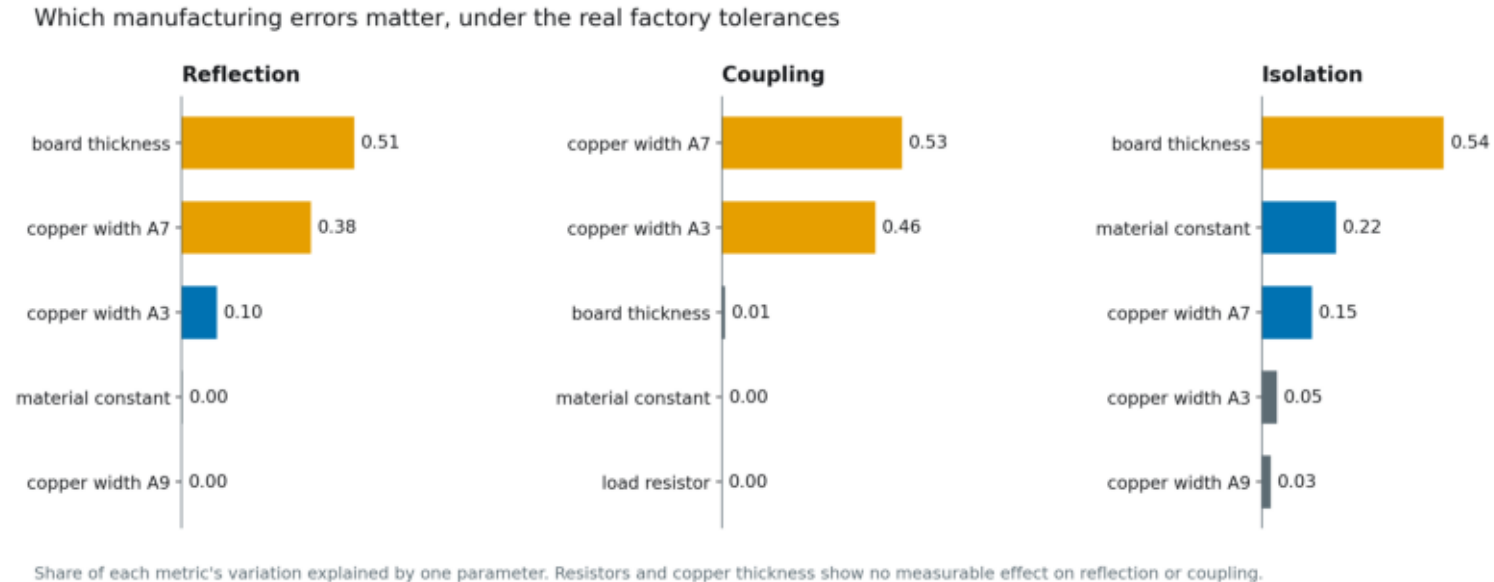
THE CORRECTION

Compressing the response first was my own mistake. Removing it improved the winner threefold.

The recommendation rests on measurement rather than fashion. Kriging also carries a confidence estimate, which is what an active-learning loop needs and no rival here provides.

OUTPUT 1

Which manufacturing errors actually matter



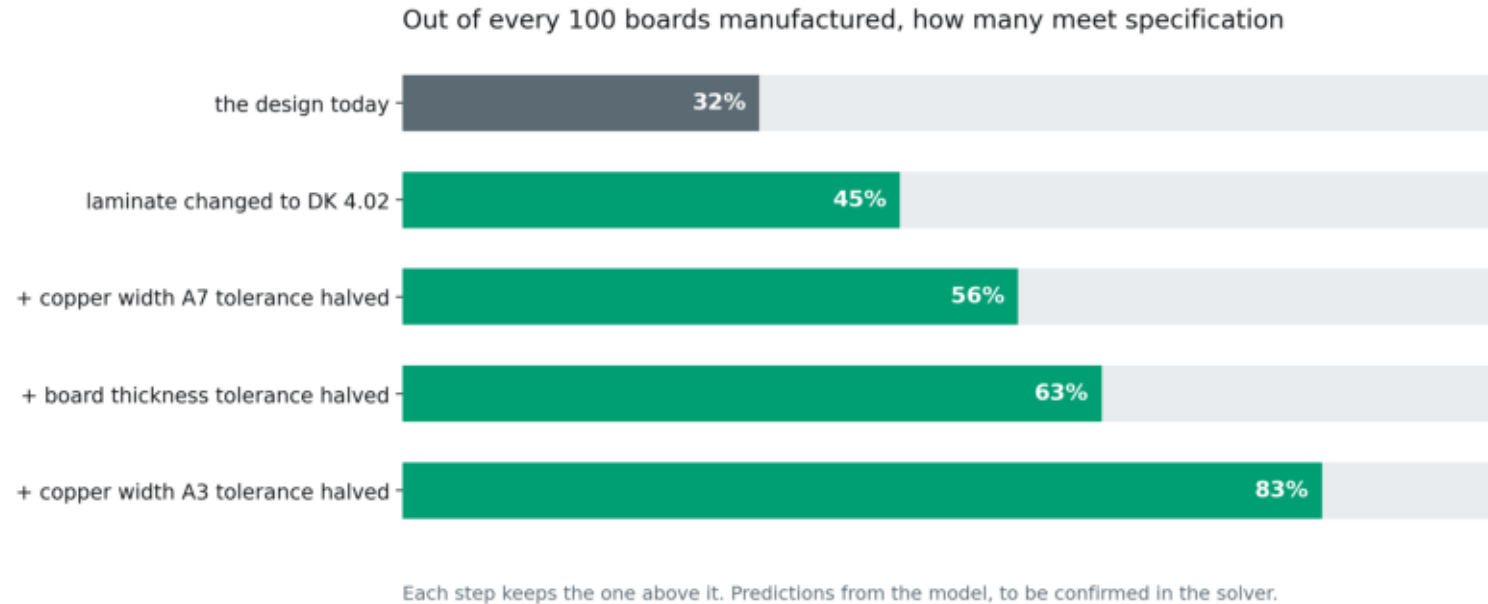
Run under the real factory tolerances Huawei supplied: 53,248 cases in 11 seconds. On the solver that is 3 years of simulation, run one after another.

Board thickness and copper width A7 still explain 89 percent of reflection. For isolation, the material constant fell from first to second place once its real 2 percent tolerance replaced the wider range in the dataset.

Resistors and copper thickness show no measurable effect on reflection or coupling, even at their full real tolerance.

OUTPUT 2

How many boards pass, and how to raise it



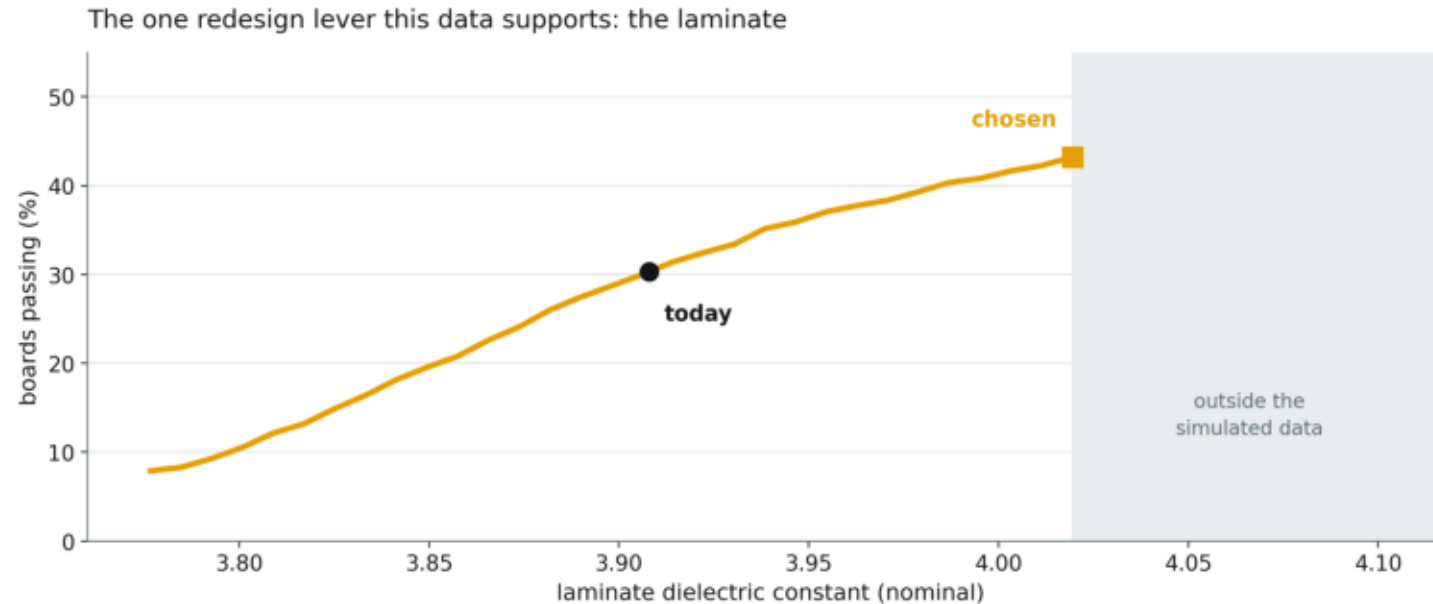
A hundred thousand virtual boards under the real tolerances took 16 seconds, against almost 6 years on the solver. Today 32 percent pass. The model agreed with the real simulator on pass or fail for 93 percent of designs it never saw.

Changing the laminate lifts that to 45 percent. Halving the tolerance on two copper widths and the board thickness lifts it to 83 percent. Each step keeps the one before it.

These are predictions. The pass rate also depends on whether factory scatter is uniform or bell-shaped: 32 percent or 70 percent for the same design.

OUTPUT 3

The redesign: one lever, and where it runs out



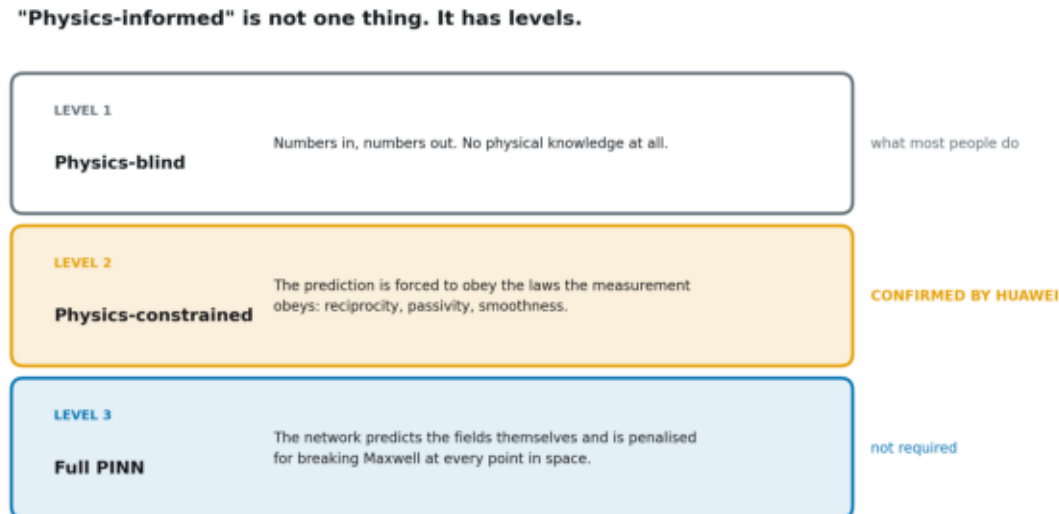
The search tried 1.9 million designs in 5 minutes. On the solver that is 110 years. Only the laminate and copper thickness could move, since the data covers every other parameter at just its own tolerance.

Raising the laminate dielectric constant from 3.91 to 4.02 cuts isolation failures from 24 to 3 percent. Copper thickness changed the pass rate by barely one point.

The best value sits at the edge of the simulated data. Simulating laminates above 4.02 is the next job, and exactly what active learning is for.

CONFIRMED

The physics this project uses counts, and Huawei said so



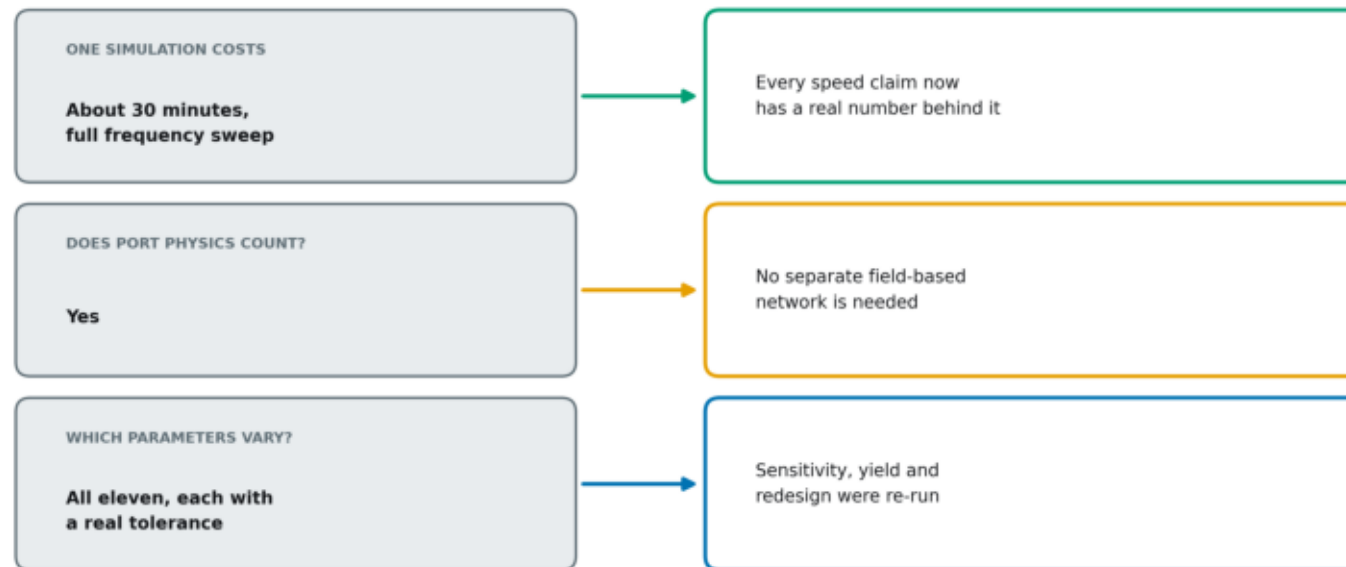
A full physics-informed network is penalised for breaking Maxwell equations at points in space. That needs the fields throughout the structure, which the supplied file does not contain.

This project enforces the laws the port measurement obeys instead: reciprocity by construction, passivity checked on every prediction.

Asked directly, Huawei confirmed that this counts as physics-informed. No separate field-based network is needed.

What Huawei answered, and what it changed

What Huawei answered, and what it changed



The 30-minute cost turns every speed claim from an expression into a number a jury can check.

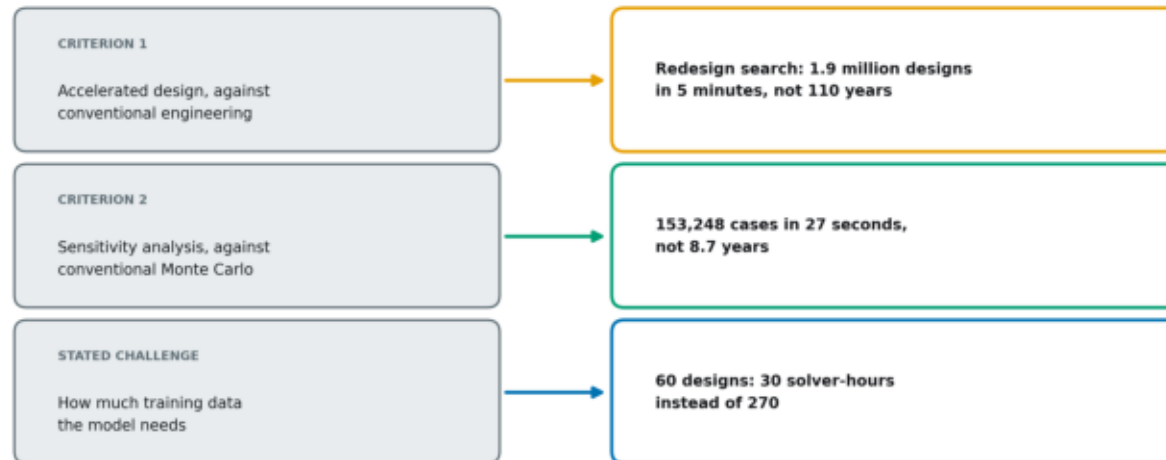
The yes on physics removes the biggest risk this project carried.

The tolerance table changed two conclusions, and both are reported.

What the marking scheme rewards, and what answers it

What the marking scheme asks, and what answers it

At Huawei's 30 minutes per simulation, run one after another.



Both criteria compare your cost with the conventional cost. At 30 minutes a simulation, both ratios are now real numbers.

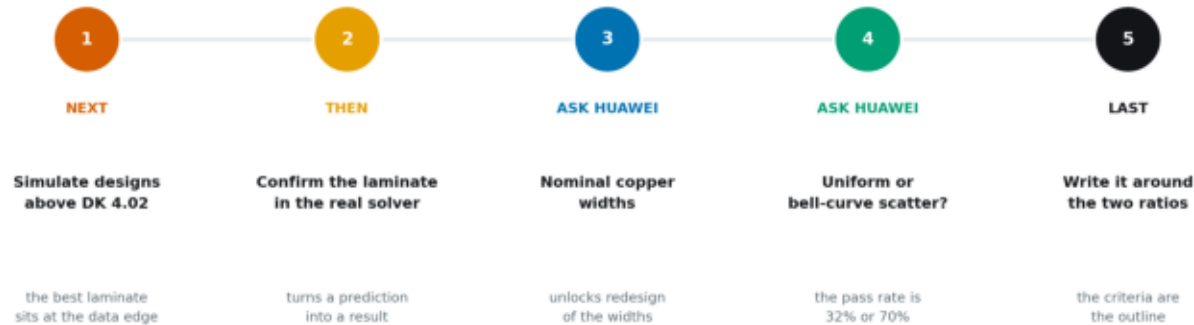
All times are serial. Even with 100 simulations running at once, the sensitivity study alone would take 11 days.

Quote time units, not a multiplier. Years against minutes reads as credible where a factor of millions does not.

THE PLAN

What to do next, in order

What to do next, in order



The first two steps turn the redesign from a prediction into a result.
The two questions for Huawei would lift the last limits on what the data can say.

The last step is not writing up what you did. It is writing the submission around the two ratios the jury scores, and letting the evidence fall under whichever ratio it supports.